

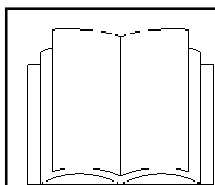


Operator's manual for Attachment



Sand and Salt Spreader

Product number:	200-series loaders:	A21114
	300-700-series loaders:	A2947



Please read this manual carefully before using the equipment, and follow all instructions.

Keep this manual for later reference.

Manufacturer:

AVANT[®]
AVANT TECNO OY
e-mail: sales@avanttecno.com

Ylötie 1
33470 Ylöjärvi
FINLAND
Tel. +358 3 347 8800
Fax +358 3 348 5511

www.avanttecno.com

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Appendix: EC Declaration of Conformity

This manual uses the following warning symbols to indicate factors that must be taken into account to reduce the risk of personal injury or property damage:

	This safety symbol refers to important safety information in this manual. It warns of an immediate hazard that could cause serious personal injury or property damage. Read the warning text accompanying the symbol carefully and ensure that other operators are also familiar with the warnings, since personal safety is at stake.
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This notice symbol indicates information about the correct operation and maintenance of the equipment.

Failure to observe the instructions accompanying the symbol can lead to equipment breakdown or other property damage.

In addition to the safety instructions included in this manual, you must observe all occupational safety regulations, local laws and other regulations concerning the use of the equipment, particularly the use of the equipment on road traffic. Due to continuous product development, some of the details in this manual may differ from your equipment. We reserve the right to change the contents of the manual without notification. This manual is a translation of the original Finnish version.

1. Foreword

Avant Tecno Oy would like to thank you for your purchase of this AVANT attachment. It has been designed and manufactured on the basis of years of product development and experience. By familiarising yourself with this manual and following the instructions, you ascertain your safety and ensure the reliable operation and long service life of the equipment. Read the instructions carefully before starting to use the equipment or performing maintenance.

With these instructions, even an inexperienced user can use the attachment and loader safely. The manual includes important instructions for experienced AVANT operators as well. Ensure that all persons using the loader have received proper guidance and familiarised themselves with the manual of the loader and attachment and all safety instructions before using the equipment. Keep this manual at hand throughout the service life of the equipment. If the manual is lost or damaged, you can request a new one from the retailer or manufacturer. If you sell or transfer the equipment, be sure to hand over this manual to the new owner.

2. Designed purpose of use

This AVANT sand / salt spreaders are attachments suitable for use with AVANT multi purpose loaders, for fast and efficient spreading of materials, which are suitable for sanding. Spreaders are powered by a hydraulic motor and they are equipped with a mixer axle, which prevents clogging up of material being spread.

Front mounted spreaders are easy to operate accurately also at narrow areas. The spreader model A2947 can be filled also by loading with the loader boom, working in a similar way as with a normal bucket. Also available is a spreader model designed to be fitted to the rear lift device available on some loader models. This manual applies only to the two front mounted models.

The attachments have been designed to require as little maintenance as possible. Regular maintenance tasks can be performed by the operator. Appropriate protective equipment and original spare parts must be used for the maintenance. Demanding repair and maintenance operations are to be left for professional maintenance. Familiarise yourself with the manual's instructions regarding service and maintenance. Please contact your AVANT retailer if you have additional questions about the operation or maintenance of the equipment or if you require spare parts or maintenance services.

3. Safety instructions for using the spreader

Please bear in mind that safety is the result of several factors. The loader-attachment combination is highly powerful, and improper or careless use or maintenance may cause serious personal injury or property damage. Due to this, you should carefully familiarise yourself with the manuals of both the loader and the attachment before starting operation. Do not use the attachment if you have not completely familiarised yourself with its operation and the related hazards. The purpose of these instructions is to help you to:

- use the equipment in a safe and efficient manner
- observe and prevent any hazardous situations
- keep the equipment intact and ensure a long service life



- The sand / salt spreader is to be used by one operator at a time. Do not approach or allow others to approach the area where the spreader is in operation.
- When mounting the spreader to the loader, ensure that the locking pins of the loader's quick attach plate are **in the lower position** and that they have **locked** the attachment **to the loader**.
- Never leave the driver's seat when the spreader is running. Keep a safety distance of at least one meter when the spreader is running. If you do not load the spreader directly form a pile with the loader, turn off the loader engine for loading.



- Ensure that the unattached spreader is properly supported. Unmount the attachment with care. Ensure that the spreader is properly supported during all inspection and maintenance procedures.
- Inspections, cleaning and removing of blockages must be done carefully.
- Do not remove the safety grille from the spreader. Do not modify the structure of the attachment or the loader in a way that would affect safety.
- Pay attention to the surroundings and any other persons and machines moving in the vicinity. Pay attention to the contours of the terrain and as well as other hazards, such as branches and trees reaching the driver's area. On slippery areas drive calmly and deliberately.
- Always transport the spreader as low and close to the machine as possible to keep the centre of gravity low.
- Always shut down the loader engine first before filling the spreader by hand or going near the spreader. Never perform any maintenance or adjustments while the loader engine is running.
- Never go under a raised attachment.
- Make sure not to tilt or raise the spreader in a way that material could spill from the spreader.
- Check the spreader axles regularly. Read the instructions on maintenance and servicing and only use original spare parts.
- Also read the safety instructions for the loader in the loader manual.



Familiarise yourself with the loader controls in a safe area. Pay particular attention to the safe stopping of the attachment and the machine. Note that the load at the front and slippery surfaces may alter the driving characteristics of the loader.



Note that the loader boom moves down and may be tilted even if the engine has been stopped. Never go under a raised boom or spreader component.

Remember to wear proper protective clothing:



- Do not wear loose clothes or gloves, long scarves, necklaces or jewellery, which might tangle with the moving parts of the spreader. Use a hat or other suitable means to prevent long hair from entering the machine.



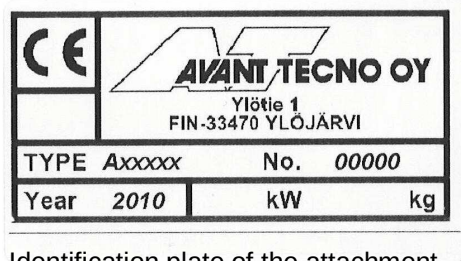
- The noise level in the driver's seat may exceed 85 dB(A) depending on loader model and working conditions. Wear hearing protection while working with the loader.
- The use of safety boots is recommended while working with the loader.
- Wear protective gloves when handling hydraulic hoses.

4. Sand / salt spreader specifications

Listed below are the labels and markings, which must be visible on the equipment. Replace any warning label, that has become unclear or has detached completely. New labels are available via your retailer or contact information provided on the cover.



The warning labels contain important safety information. Replace damaged or missing warning labels with new ones.



Identification plate of the attachment

- A46771 – Read instructions before use.
- A46772 – Do not go under a raised attachment, stay away from raised equipment.
- A46798 – Risk of getting squeezed between moving components; do not use without all covers in place.
- A46801 – Entanglement hazard, stay away from moving parts.



A46771



A46772



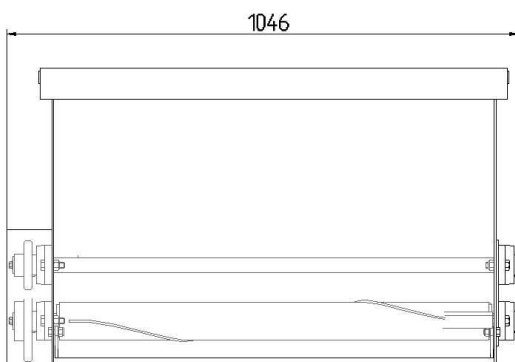
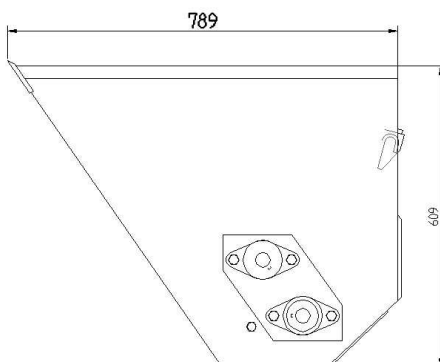
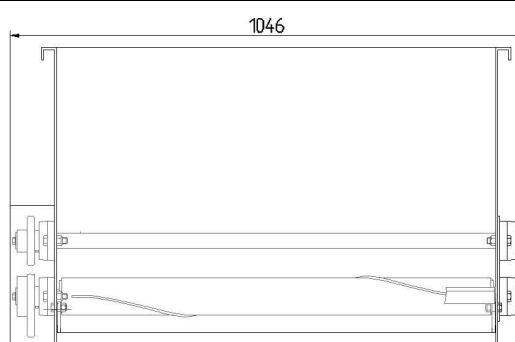
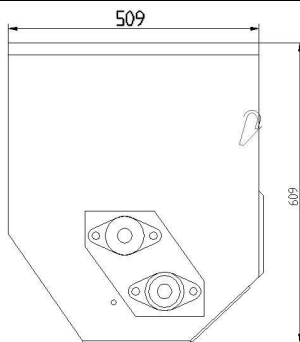
A46798



A46801

Table 1 – Sand / Salt Spreader specifications

Product number:	A21114	A2947
Working width:	1 000 mm	900 mm
Volume:	160 l	200 l
Weight (empty):	135 kg	170 kg
Maximum input of hydraulic energy:	22 MPa (220 bar), 40 l/min	22,5 MPa (225 bar), 60 l/min
Suitable AVANT loader:	200	300-700



5. Assembly of the spreader

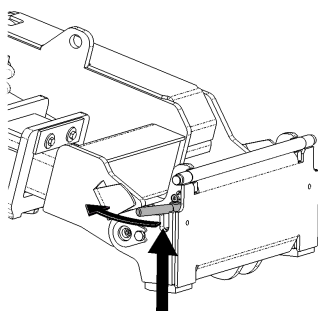
The spreaders are delivered in working condition. Mounting the attachment to the loader is quick and easy, but it must be done carefully. An attachment that has not been locked may come off the loader and cause hazardous situations. In the worst case, the attachment may slide onto the driver along a raised boom. For this reason, the attachment must under no circumstances be raised to a height of over one metre when it has not been locked. To prevent hazardous situations, always follow the attachment mounting instructions provided in the following pages. Also remember the safety instructions in Section 3 and the instructions provided in the loader manual.

5.1. Mounting the spreader to the loader

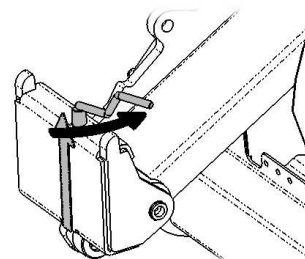
These instructions are applicable to mounting the spreader to the boom quick attach plate of AVANT loaders that have standard quick attach locking. If your machine is equipped with a hydraulic locking and/or serial quick attach system, refer to the relevant manuals for additional mounting instructions.

The attachment is mounted to the loader boom by using the quick attach plate and the counterpart on the attachment. The locking is semi-automatic; when the boom front plate is tilted towards the loader, the locking pins are released as they hit the pin guides. We recommend that you lock the pins manually, as you attach the hydraulic hoses. The attachment is mounted on the loader as follows:

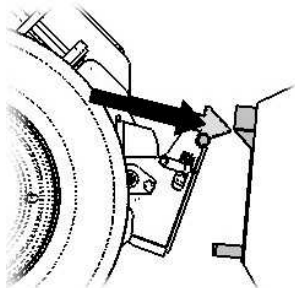
Step 1:



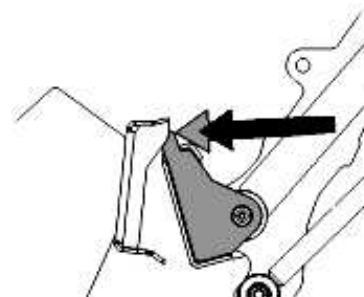
- Lift the quick attach plate locking pins up and turn them backwards into the slot so that they are locked in the upper position.
- Ensure that the hydraulic hoses are not in the way during installation.



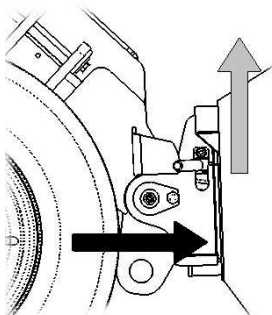
Step 2:



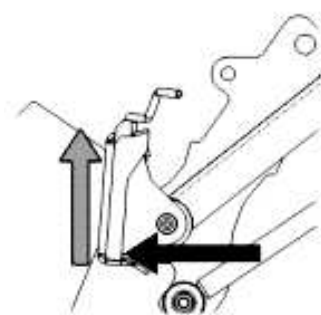
- Turn the quick attach plate hydraulically to an obliquely forward position. The angle between the loader front plate and attachment coupling should be about as shown in the adjacent picture.
- Drive the loader onto the attachment. If your loader is equipped with a telescopic boom, you can utilise this.



Step 3:

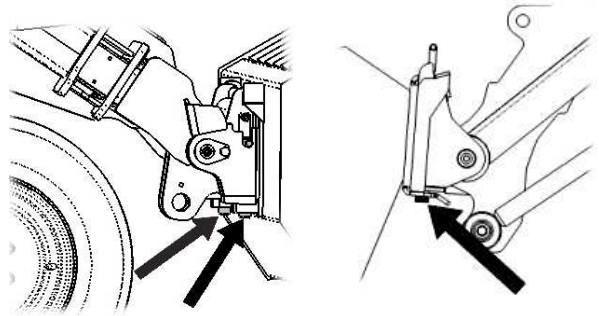


- Turn the boom control lever left to turn the bottom section of the quick attach plate onto the attachment.
- Lift the boom slightly – pull the boom control lever backward to raise the attachment off the ground.
- Turn the boom control lever further left so that the automatic locking of the attachment triggers the locking pins.





Before moving or raising the attachment, ensure that the locking pins are in the lower position and come through the fasteners on both sides. Unlocked attachment may fall on the boom or operator and cause injuries.



Do not use the automatic locking of the locking pins at a height of over half a metre. If the locking pins do not return to the normal position when the front plate is tilted, do not raise the attachment any more. Lower the attachment on the ground and secure the locking manually.

5.2. Connecting and disconnecting the hydraulic fittings

The three hydraulic fittings of the spreader must all be connected to the corresponding fittings on the loader. Before connecting or disconnecting the fittings, shut down the loader engine and move the auxiliary hydraulics lever on the loader to its extreme positions a few times. This way, there is no remaining pressure in the system and the fittings are easy to connect.

To connect and disconnect the fittings, move the collar at the end of the female fitting. For correct rotating direction of the spreader, the hoses must be connected to the loader so that the fittings equipped with coloured caps are connected to each other



Always ensure that all three fittings are properly connected to the corresponding fittings on the loader. If any of the hoses is not properly connected, the hydraulic motor of the spreader may be damaged.



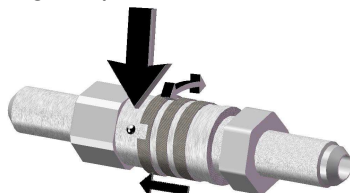
Never connect or disconnect quick couplings or other hydraulic components while the system is under pressure. High-pressure fluid ejection may cause serious injuries.



When removing the attachment, always disconnect the hydraulic fittings before detaching the quick attach plate, to prevent hose damage and any oil spills. Reinstall the protective caps on the fittings to prevent impurities from entering the hydraulic system.

Keep the fittings as clean as possible; use the protective caps for the quick couplings on both the attachment and the loader. Dirt, ice, etc. may make using the fittings significantly more difficult. Note that the protective caps on the loader and the attachment can be fastened to each other during operation to reduce the accumulation of dirt. Never leave the hoses hanging on the ground; place them on top of the equipment.

In a 200-series machine, the hose can be connected by pushing the fitting in the counterpart. When connecting and disconnecting the 200-series hydraulics quick couplings, you should pay attention to their locking. The locking prevents the female and male fittings from being disconnected from each other by accident when the attachment moves. The fitting collar may also turn by accident into the locking position, making it impossible to connect or disconnect the fittings.



In the 200-series equipment, there is a small groove in the collar of all female fittings and a small ball below the collar. When the fittings need to be disconnected, the above-mentioned groove and ball need to be aligned. Then, the grooved collar of the female fitting is pulled downwards to disconnect the fittings from each other.

6. Instructions for use

Work can be started once the sand / salt spreader is mounted according to the instructions and the hydraulic fittings are connected. Check the attachment and the operating environment once more before starting to work. Quick inspection of the sand spreader prior use is part of ensuring safety and the best performance of the equipment.

- Check that the spreader is in order and that the spreading and mixer axles can rotate freely. Check that there is no string or other foreign material coiled around the axles.
- Remove larger sand clogs or split them to smaller pieces for even sanding results.
- Check that all obstacles, including any possibly hidden ones, have been removed from the working area before operation. Remember correct working methods and avoid getting down unnecessarily from the loader.
- Check regularly to ensure there are no hydraulic leaks.
- Ensure that the loader and the attachment are used in a safe and intended manner. Do not allow children to operate the machine.
- Operate the attachment from the driver's seat only.



Due to the operating principle of the spreaders, they can be used to their intended rotating direction only. The rubber lip plate and the sweeping plate (see section 7.2) may get damaged, if the spreader is not used properly. If material does not come out from the spreader or it comes out poorly, first check the direction of rotation and then the settings of the spreader (see section 6.3).



Never put your hand inside the spreader or try to reach the axles by hand. To remove clogs or lumps, use an aid to remove the clog. For example, use a brush handle or some other suitable object. Watch out that clothes do not get tangled with the spreader axles. Always turn off the spreader before descending from the loader.

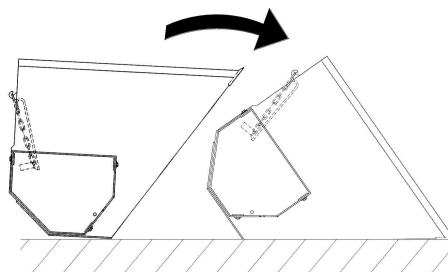


Before using the spreader, check that it is intact and that all protective covers and the protective grille inside the spreader are in place. Do not use the attachment if any of the protective covers is missing.

To ensure the stability of the spreader during storage, use it until it is empty or empty the spreader on purpose before demounting the attachment. The spreader should be stored so that the longest side is against the ground surface, as shown in figure below, not in the upright position.



Unattached spreader may tip over; keep a safe distance. Ensure the stability by placing the attachment firmly on the ground on a stable position.



6.1. Filling the spreader

Lifting material from a pile requires the most power from the loader. To ensure the best performance of equipment, the correct working method must be observed especially when using the sand / salt spreaders. Filling the large bucket by ramming straight into a pile of heavy material and then trying to lift or tilt the bucket is not possible with any loader type. The loading is most effective when only the tip of the bucket is pushed into the pile, and then by simultaneous driving, lifting and tilting the equipment, material is cut into the bucket. Use high engine revolutions and read the instructions for correct use of the drive pedal from the loader manual.



Ensure that the loader can handle the load intended to be moved. Notice the risk of the loader tipping forward when material is being lifted from a high level. Do not drag the filled attachment from a high level.

Fill the sand / salt spreader only when it is firmly attached to the loader. If you fill the spreader with some other equipment or by hand, stop the spreader before going near it. Lower the equipment on the ground and observe that it might tilt forward, even if attached. Avoid staying directly in front of the spreader.

The spreaders work the best with crushed rock or gravel that consists of homogenous max. 6 mm pebbles and which is intended for sanding. Very fine sand may not proceed through the sand spreader rollers rapidly enough, which results in inadequate sanding speed, or unsuccessful spreading. Clogged or frozen gravel or dense sand may block the spreader regardless of the mixing axle on the spreader. When finished sanding, do not leave material in the spreader, as the material may clod or freeze and using the spreader the next time may be more difficult. Moreover, material left in the spreader may increase the risk of the unattached spreader getting tipped over.

6.2. Using the sand / salt spreader

The spreader is operated with the auxiliary hydraulics control lever on the loader. The spreader is stopped by releasing the control lever or by shutting down the loader, and the attachment will stop. There is no separate stop switch on the spreader. If your loader is equipped with the optional electric joystick (available on 600 and 700 series loaders), the precise control of the spreading is easy by using the electric switches on the joystick.

When using the spreader, keep it in an upright position at a suitable height from the ground, so that the lower edge of the spreader or the spreading axles never hit the ground, or any obstacles, at any conditions. The rubber lip plate under the spreader or the spreading axle itself may get damaged if hit, resulting in uneven spreading of material, or leading to more serious damage to the spreader. See section 7.2 on page 10 for more information about the rubber lip and the sweep plate.

6.3. Adjusting the spreader

The spreader is used with one or two loader auxiliary hydraulic pumps, depending on the model of the loader, as well as the driving speed, the settings of the spreader and the type of the material being spread. Generally more fine-grained material will require greater auxiliary hydraulic flow. The adjustable lower plate should be set according the type of the material being spread. The tilting of the spreader also slightly affects the amount of spreaded material.

6.3.1. Adjustable lower plate

The lower plate of the spreader is adjusted with a pair of chains, which are attached to the spreader frame. The adjusting chains are fitted to the lower plate through springs. When more sand is wanted, or a larger grain material is being used, the springs should be loosened. Adjust both of the chains equally and by little at a time to find the suitable setting. Observe that releasing the chains completely may lead to the sand being drained from the spreader.



The springs of the spring-adjusted spreader are tensioned. Grasp the chain with a secure grip by both hands and adjust the chains with care, adjusting one chain at a time.

The setting of the lower plate is adjusted according to the desired amount of material, driving style and the loader model. The best adjustments according to loader and specific material type may be found by testing the setting and adjusting as necessary.

7. Maintenance and servicing

Continuous maintenance of the sand spreader includes regular cleaning and lubrication of the equipment. The chain drive should be inspected when the lubricating points are lubricated and adjusted when needed. Do not store the attachment directly against the ground; place it on blocks of wood or on a pallet, for example, and ensure the stability of the equipment. Cover the sand spreader in order to avoid snow and ice from blocking the spreader.



Ensure that the sand spreader is properly supported during storage and all inspection and maintenance procedures. Use the service support on the loader and ensure sufficient additional protection to prevent the tilting of the equipment. Falling tilted sand spreader may cause serious injuries.

Check the condition of the hydraulic hoses and components when the engine has been turned off and the pressure has been relieved. Repair all leaks immediately after detecting them; a small leak can quickly grow into a big one. Do not use the equipment, if you have discovered a leak in the hydraulic system. Leaking hydraulic fluid may penetrate skin and cause serious injuries. Hydraulic fluids are also harmful to the environment.

Check hoses visually for cracks or abrasions. Monitor the wearing of the hoses and stop the use, if the surface layer of any hose has worn off. If there are signs of leaking, to check the component, hold up a piece of cardboard in the area where a leak is suspected. Finding any of these faults means that the hydraulic hose or component must be replaced. Spare parts are available from your nearest AVANT retailer or service point.



Never tighten a hydraulic fitting under pressure, since the fitting may break and the released oil may cause serious injuries. Do not use the equipment, if you have discovered a leak in the hydraulic system.



7.1. Cleaning and lubricating the sand spreader

Clean the attachment thoroughly before long storage periods, especially when a material that includes salt has been used. You may use a pressure washer and a mild detergent to wash the equipment. Avoid spraying directly at the hydraulic components or using solvents when washing. In addition, mind the labels on the attachment. Touch up the paint where needed to prevent rust damage. Lubricate the lubrication points after washing.

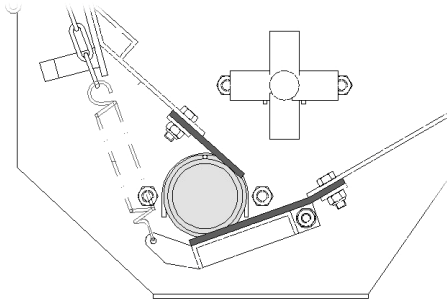
There are 4 lubricating points on the spreader, which can be lubricated with a standard grease press. The lubricating points are located at the axle bearings. The lubricating points at the free ends of the axles are directly accessible. To lubricate the other two lubricating points the protective cover must be opened. Replace the protective covers before using the equipment.



Lubricant must be added approximately every 5 to 10 operating hours. Clean the end of the nipple before greasing. Only add the grease amount of a couple of grease press pushes at a time.

7.2. The rubber lip plate and the sweeping plate

The operation of the spreaders is based on the two rubber plates leaning against the spreading axle, which makes controlled spreading of the material possible. If the spreading result is uneven after cleaning and adjusting the equipment, the condition of the rubber plates, shown in the figures below, must be checked. The edge of the plates should be even, and they should be adjusted to lean lightly against the axle. Please contact your nearest AVANT service if needed.



Spreader models A21114 and A2947
Rubber plates and spreading shaft

7.3. Checking and adjusting the chain drive

The roller chain transmission between the hydraulic motor and the mixing and spreading axles should be inspected whenever the protective cover is opened for lubricating the bearings. Clean the chain housing and lubricate the chain with suitable type of oil. The tightness of the chain can be checked by pushing the chain between the sprockets with moderate force. The clearance of the chain should be about 0.5 – 1 cm. If necessary, the chain is adjusted by moving the hydraulic motor until the chain is at correct tightness. Tighten any loosened bolts and fit the protective cover back to its place.

8. Warranty terms

Avant Tecno Oy grants a warranty of one year (12 months) from the date of purchase for the attachment it manufactures.

The warranty covers repair costs as follows:

- Work costs are covered, if the repair is not performed at the factory.
- The factory replaces any defective components or consumables.

The factory may reimburse the price of components purchased by the customer in special cases that have been agreed in advance.

The warranty does not cover:

- Normal maintenance work or parts and consumables required for it.
- Damages caused by unusual operating conditions or ways of use, negligence, structural changes made without the consent of Avant Tecno Oy, use of non-original parts or lack of maintenance.
- Consequences of a defect, such as interruption of work or other possible additional damages.
- Travel and/or freight costs caused by the repair.

EY-vaatimustenmukaisuusvakuutus
EG-försäkran om överensstämmelse
EC Declaration of Conformity
EG-Konformitätserklärung



Valmistaja / Tillverkare / Manufacturer / Hersteller:

AVANT TECNO OY

Osoite / Adress / Address / Adresse:

Ylötie 1

33470 YLÖJÄRVI, FINLAND

Vakuutamme täten, että alla mainitut tuotteet täyttävät konedirektiivin turvallisuus- ja terveystvaatimukset (direktiivi 2006/42/EY muutoksineen). Seuraavia yhdenmukaistettuja standardeja on sovellettu:

SFS-EN ISO 12100, SFS-EN ISO 4413, SFS-EN 13021

Vi försäkrar härmed att nedan beskrivna produkter överensstämmer med hälso- och säkerhetskrav i EG-maskindirektiv (EG-direktiv 2006/42/EG som ändrat). Följande harmoniserade standarder har tillämpats:

SFS-EN ISO 12100, SFS-EN ISO 4413, SFS-EN 13021

We hereby declare that the products listed below are in conformity with the provisions of the Machinery Directive (directive 2006/42/EC as amended). The following harmonized standards have been applied:

SFS-EN ISO 12100, SFS-EN ISO 4413, SFS-EN 13021

Wir erklären hiermit, dass die nachstehend aufgeführte Maschine konform ist mit den Bestimmungen der EG-Maschinenrichtlinie (EG-Richtlinie 2006/42/EG mit Änderungen). Die folgende harmonisierte Normen wurden angewandt:

SFS-EN ISO 12100, SFS-EN ISO 4413, SFS-EN 13021

Kuvaus: Avant-kuormainten kanssa käytettävä hiekoitin
 Beskrivning: Sandspridare; arbetsredskap för Avant lastare
 Description: Sand spreader; attachment for Avant loaders
 Beschreibung: Sandstreuer; Anbaugerät für Avant Radlader

Mallit / Modeller / Models / Modellen:

Avant	200-series	300-700-series
Hiekoitin Sandspridare Sand spreader Sandstreuer	A21114	A2947

Allekirjoitus / Namnteckning / Signature / Unterschrift:

26.4.2011 Ylöjärvi, Finland

Risto Käkälä,
 Toimitusjohtaja / Verkställande direktör /
 Managing Director / Geschäftsführer





AVANT[®]